

1 policy has only one slave in the bond active. A different slave becomes active only if the active slave fails. The bond's MAC address is externally visible on only one port (network adapter), to avoid confusing the switch.

- **miimon:** This integer value is used for MII link monitoring. The default is zero, which indicates that link monitoring is off.
- **downdelay:** This integer is the number of milliseconds by which disabling of a link is delayed, once link failure is detected.
- **updelay:** The opposite of *downdelay*. This is again an integer value, and must be a multiple of *miimon*.
- **arp_interval:** An integer value in milliseconds, specifying the frequency of ARP monitoring. The default value is zero, which indicates that ARP monitoring is disabled.
- **arp_ip_target:** An IP address to use when *arp_interval* is > 0. This is the target of the ARP request sent to determine the health of the link to the target.
- **lacp_rate:** The rate at which the link partner is asked to transmit LACPDU packets in 802.3ad mode. Possible values are *slow* or *0* (request partner to transmit LACPDUs every 30 seconds—the default option) and *fast* or *1* (transmits LACPDUs every second).

Configuring bonding manually

If your distribution's network initialisation package is not aware of bonding, then you can fall back to the manual method. In general, this method is to place the bonding module parameters into */etc/modules.conf* or */etc/modprobe.conf* (as appropriate for the installed distro), then add *modprobe* and/or *ifenslave* commands to the system's global init script. The name of the global init script differs; for *sysconfig*, it is */etc/init.d/boot.local*, and for init scripts, it is */etc/rc.local*. Let's go into greater detail about creating the *bond0* interface using the manual method.

Edit the system's global init script and add these entries, replacing the bonding module parameters and network configuration (IP address, netmask) with the appropriate values for your network:

```
modprobe bonding mode=balance-alb miimon=100
ifconfig bond0 192.168.1.1 netmask 255.255.255.0 up
ifenslave bond0 eth0
ifenslave bond0 eth1
```

To shut down the bonding devices, it is necessary to first mark the bonding device itself as being down, and then remove the appropriate device driver modules. For our example above, you can do the following:

```
[root@bastion ~]# ifconfig bond0 down
[root@bastion ~]# rmmod bonding
```

To remove the bonding configuration in init scripts, simply delete the *ifcfg-bond0* file and restore the *ifcfg-ethX*

files; restart the network services, and you are done.

Verify if everything is working by executing the following commands:

```
[root@bastion ~]# cat /proc/net/bonding/bond0
Ethernet Channel Bonding Driver: v3.5.0 (November 4, 2008)
Bonding Mode: load balancing (round-robin)
MII Status: up
MII Polling Interval (ms): 0
Up Delay (ms): 0
Down Delay (ms): 0
Slave Interface: eth2
MII Status: up
Link Failure Count: 0
Permanent HW addr: 00:0c:29:36:4d:bc
Slave Interface: eth3
MII Status: up
Link Failure Count: 0
Permanent HW addr: 00:0c:29:36:4d:c6

[root@bastion ~]# ifconfig -a
bond0  Link encap:Ethernet  HWaddr 00:0C:29:36:4D:BC
        inet addr:192.168.140.251  Bcast:192.168.140.255
        Mask:255.255.255.0
        inet6 addr: fe80::20c:29ff:fe36:4d0c/64 Scope:Link
        UP BROADCAST RUNNING MASTER MULTICAST  MTU:1500  Metric:1
        RX packets:1968452 errors:0 dropped:0 overruns:0 frame:0
        TX packets:1973388 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:0
        RX bytes:212172292 (202.3 MiB)  TX bytes:118232600 (112.7 MiB)

eth2  Link encap:Ethernet  HWaddr 00:0C:29:36:4D:BC
        UP BROADCAST RUNNING SLAVE MULTICAST  MTU:1500  Metric:1
        RX packets:1849261 errors:0 dropped:0 overruns:0 frame:0
        TX packets:1115402 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:1000
        RX bytes:155045472 (147.8 MiB)  TX bytes:71836050 (68.5 MiB)
        Interrupt:19 Base address:0x2000

eth3  Link encap:Ethernet  HWaddr 00:0C:29:36:4D:BC
        UP BROADCAST RUNNING SLAVE MULTICAST  MTU:1500  Metric:1
        RX packets:919192 errors:0 dropped:0 overruns:0 frame:0
        TX packets:857988 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:1000
        RX bytes:57126026 (54.4 MiB)  TX bytes:46396128 (44.2 MiB)
        Interrupt:19 Base address:0x2000
```

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